6

Reinterpreting

As a literature develops, a conventional wisdom often enshrines a particular theoretical interpretation of some empirical findings. This might be a sign of success—agreement between empirical findings and theoretical implications provides evidence for similarity between the theoretical model and the target and thus evidence the interpretation is correct. But we should always ask about other plausible interpretations. Offering such reinterpretations often productively moves a literature forward.

To be a reasonable interpretation of a set of empirical findings, a model must satisfy two conditions. First, it must be plausibly similar to the target. Second, its implications must agree with the empirical findings. Suppose there is already one such interpretation. A scholar reinterprets by proposing a different model, embodying different mechanisms, that is also a reasonable interpretation of those same empirical findings.[1] Looked at through the lens provided by our framework, this means the scholar could plug in the new theoretical model on the left-hand side of Figure 2.1 while maintaining commensurability, agreement between the theoretical implications and empirical findings, and the plausibility of similarity.

Some additional terminology can help clarify what's going on here. Call an implication shared by two theoretical models a *common implication* of those two models. A reinterpretation involves pointing out that some existing empirical findings actually agree with common implications of a model embodying the old mechanism and a model embodying the new mechanism. If the two models are both plausibly similar to the target, then that agreement does not favor one interpretation over the other.

1. The requirement that the new model embody different mechanisms is to exclude trivial cases in which the two models differ only in auxiliary features.

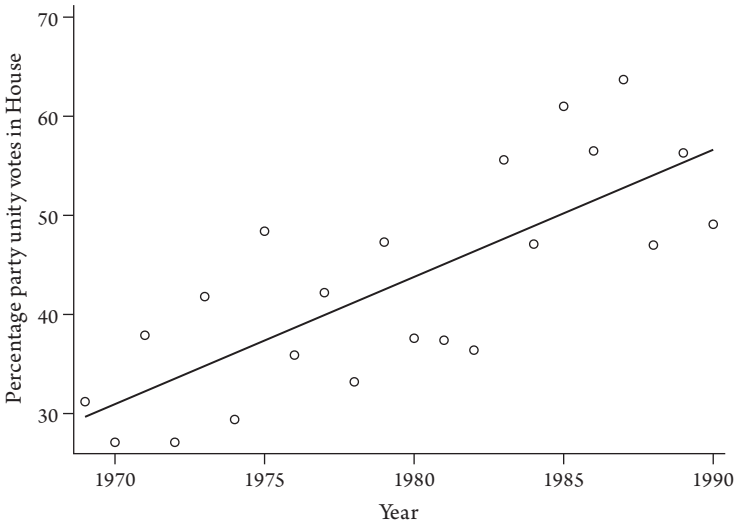

FIGURE 6.1. History of party unity voting, 1969–1990.

Importantly, a reinterpretation need not assert that the previously posited theoretical mechanism is absent, nor need it assert that the newly proposed one is present. It is instead an argument about the evidence on offer. Because that evidence is consistent with more than one mechanism, a reasonable interlocutor is not forced to accept one interpretation over the other.

In this chapter, we explore these ideas through three substantive literatures. The first concerns the influence of parties on the roll-call votes of members of the US Congress. The second concerns mobilization and recruitment for terrorist groups. And the third, which we alluded to briefly in Chapter 1, concerns the effect of natural disasters on the electoral fortunes of incumbent leaders.

6.1 Party Effects in Congress

During the 1980s, students of the US Congress observed a resurgence in party unity voting—the proportion of roll-call votes in which a majority of Democrats vote one way and a majority of Republicans vote the other. Figure 6.1 plots the history of party unity voting, revealing an upward trend, with a noticeable uptick in the eighties. Scholars including Rohde (1991) and Patterson and Caldeira (1988) interpreted this rise as evidence that political parties and their leaders were exerting increasing influence over the votes of

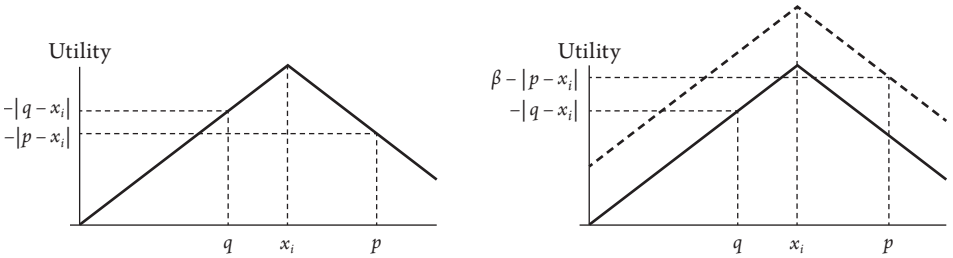

FIGURE 6.2. Increased party influence can change a legislator's vote.

rank-and-file members. A literature emerged explaining this apparent resurgence in terms of the power congressional parties wield over their members (for a review, see Rohde, 2013).

Krehbiel (1993) responded to this emerging literature with a textbook example of reinterpreting, asking "Where's the Party?" Krehbiel did not challenge the empirical claim that roll-call voting was becoming increasingly partisan. Instead, he made two points. First, an increase in party influence would matter only if legislators had some other motivation that also affected their votes. An example is close to hand. Democrats and Republicans typically have different ideologies on top of being members of different parties. Liberals tend to be Democrats, and conservatives tend to be Republicans. Second, changes to the distribution of this ideological motivation can also affect roll-call voting. Taken together, these two points make up Krehbiel's reinterpretation—the increase in party unity voting could arise from greater ideological sorting into parties.

We can represent these two mechanisms in a canonical model. To capture the party influence mechanism, assume that a legislator gets a benefit of $\beta \geq 0$ when voting as directed by the party leader. This benefit represents party-directed campaign funding, support from party leaders on other initiatives, and so on. If β is large, then legislators are highly responsive to party influence. To capture the ideological sorting mechanism, assume that bills are located in a one-dimensional policy space, and each legislator i has an ideal point x_i. Legislator i evaluates a bill with location x according to the utility function $-|x - x_i|$. An ideal point, x_i, represents the policy position this legislator would adopt without any influence from the party. This need not represent her personal policy preferences—the ideal point can also reflect electorally induced position taking.

To see Rohde's idea, consider the effect of an increase in a legislator's direct benefit from voting with the party, β. The left-hand panel of Figure 6.2

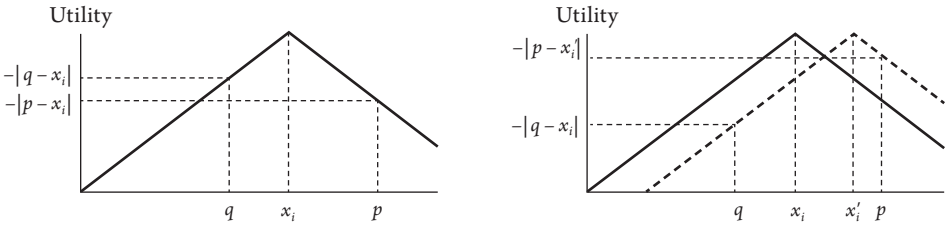

FIGURE 6.3. A shift in preferences can change a legislator's vote.

illustrates the legislator's utility function when $\beta = 0$, so there is no benefit to voting with the party. This panel shows a case in which a Republican legislator's ideal point is between the status quo policy q and a Republican party proposal p. Since the legislator's ideal point is closer to q than to p, and the legislator doesn't care at all about party pressure, the legislator votes against the proposal.

In terms of this model, Rohde's interpretation of the increased frequency of party unity voting is about a change in β, the size of the direct benefit from voting with the party leader. The right-hand panel of Figure 6.2 shows the same situation as before, but now the legislator gets benefit $\beta > 0$ if she votes for the proposal. As a result, her payoff to voting for p is given by the dashed curve, while her payoff to voting against p is still given by the solid curve. In the depicted case, the shift up by β is enough to reverse the legislator's vote.

Krehbiel's reinterpretation can also be represented in the model. Increased ideological sorting corresponds to shifting the legislator ideal points to be more in line with party proposals. Figure 6.3 starts with the same situation as in Figure 6.2. But this time the right-hand panel shows what happens if the Republican legislator's ideal point is shifted to the right (from x_i to x_i'). This shift is enough to reverse the legislator's vote, even though the direct benefit to supporting the party remains 0.

In other words, Krehbiel argues that increased party unity voting is a common implication of both increased partisan pressure and increased ideological sorting. As a result, Rohde's findings can be reasonably interpreted in terms of a model in which parties play no role, congresspeople vote based on ideology, and ideological sorting by party has increased over time.

Recall, to reinterpret an empirical finding is not necessarily to argue that the initial interpretation is wrong. Indeed, referring to Rohde's interpretation that increased party unity voting is evidence of increased party discipline,

Krehbiel writes, "such inferences are not necessarily wrong, but they are not necessarily right either" (p. 238). Krehbiel shows that Rohde's findings can be explained entirely in terms of a different theoretical mechanism. More evidence, of a different kind, is needed before we can conclude that the party influence mechanism is at work. Later in his article, Krehbiel attempts to provide evidence that the party influence mechanism is not at work in the committee assignment process, but Krehbiel's reinterpretation of Rohde's findings does not stand or fall with that exercise.

Finally, it is worth relating Krehbiel's particular reinterpretation of the party effects literature back to the framework in Chapter 2. Once we see that the party influence mechanism works alongside an ideological sorting mechanism, it becomes apparent that Rohde is implicitly holding ideologies constant in deriving his theoretical implication. Thus, commensurability of that implication and an empirical result requires a research design that also holds ideologies constant. But, as Krehbiel's argument makes clear, ideology is an omitted variable in Rohde's analysis, so commensurability fails here. The same goes for an assessment of the ideological sorting mechanism—a commensurable estimand would hold party influence fixed, asking how party-unity voting changes with changes in the distribution of ideology, all else equal. In Chapter 8 we discuss research designs that do better on commensurability for related questions.

For our purposes, it is not central that this argument can be understood in terms of omitted variables. As we will see in the rest of this chapter, theoretical reinterpretation is often possible and productive even absent an omitted-variables bias problem.

6.2 Who Becomes a Terrorist?

Scholars and policymakers share a common intuition that people often turn to political violence out of a lack of economic opportunity. Improvements in economic opportunity are thus expected to reduce mobilization for political violence. This economic opportunity-cost mechanism can be empirically assessed in a variety of ways. For instance, in Chapter 1 we discussed work examining the relationship between economic shocks and civil war, which we return to in Chapter 9. Parts of the terrorism literature take another approach, which is our focus here. The idea is to assess the opportunity-cost mechanism through a comparison of the characteristics of individuals who do and don't mobilize for armed conflict.

Krueger and Maleckova (2003) and Berrebi (2007) use this approach in studies of the individual characteristics of operatives from *Hezbollah* and *Hamas*, respectively. They show that these operatives have economic and educational statuses above the averages in their societies. For example, Berrebi reports that only 16% of Hamas operatives are poor, compared to 31% of all Palestinians. Krueger and Maleckova argue that, since those who mobilize for violence are neither poor nor poorly educated, economic opportunity must not be an important determinant of terrorist mobilization. Indeed, based on this evidence, Krueger (2007) argues, "there is not much question that poverty has little to do with terrorism."

Bueno de Mesquita (2005) offers a reinterpretation. Without disputing the empirical findings, he argues that the individual-level evidence does not entail the conclusion that economic opportunity is unimportant for understanding why individuals are willing to mobilize for political violence.

To do so, he adds organizational behavior to a model of mobilization. The idea is that two decisions must be made before someone becomes a terrorist operative—an individual decides to mobilize and a terrorist organization decides to recruit that individual. The key mechanism is screening—terrorist organizations select potential recruits based on ability, which is positively correlated both with efficacy as a terrorist operative and with economic success. (Hassan's [2001] qualitative evidence is consistent with that assumption.) The model implies that terrorist operatives will be neither poor nor ill educated, and is thus consistent with the empirical evidence. But it also implies that mobilization is responsive to economic opportunity costs.

To see the argument, let's start with a model designed to capture just the opportunity-cost mechanism that Krueger is implicitly testing and rejecting. Consider a set of individuals, each of whom has an ideological affinity with the terrorist group and so might consider mobilizing. Individual i has an underlying ability, θ_i. Each individual chooses between working in the regular economy or joining a terrorist organization. Individual i's payoff is $u_E(\theta_i)$ if she works in the economy, while her payoff is $u_T(\theta_i)$ if she joins the terrorist organization.

The parameter θ represents an individual's talents and level of education. As we illustrate in Figure 6.4, each of the functions u_E and u_T is increasing in ability; that is, individuals with higher ability expect to be more successful in the economic arena and more efficacious as terrorist operatives. The economic arena allows greater sorting according to ability, so payoffs increase faster in that sphere, as reflected in the curve labeled "Payoff from economy"

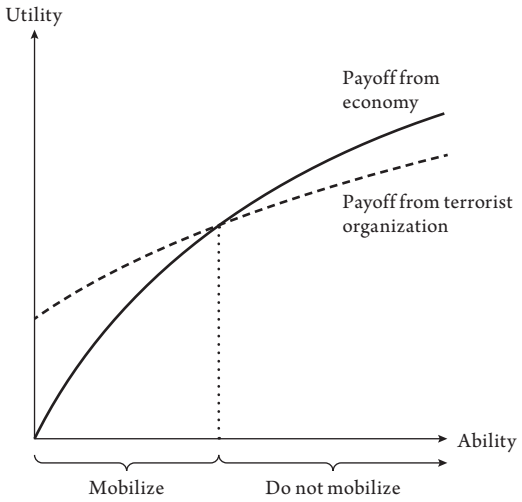

FIGURE 6.4. Terrorist mobilization decision.

being steeper than the curve labeled "Payoff from terrorist organization." An individual mobilizes for terrorism if the payoff to terrorism is greater than the payoff to economic activity. As Figure 6.4 shows, this means an individual is willing to mobilize only if her ability is below some threshold. As a result, poorer and less educated individuals, who face lower opportunity costs, are more likely to mobilize for terrorism. Krueger argues against such a model on the grounds that he does not find terrorist operatives to be poor or uneducated. Rather, he concludes, the right model has individuals choosing whether to mobilize for reasons unrelated to the economy.

But suppose we add the screening mechanism to the model in Figure 6.4—allowing terrorist organizations to choose whom to recruit from among those individuals who are willing to mobilize. There are costs and benefits to accepting a recruit. The more operatives recruited, the more attacks the organization can carry out. But accepting an operative also brings costs for the organization, for example, increased financial costs, risk of disloyalty, and so on.

Bueno de Mesquita captures these qualitative features of terrorist organizations in a variant on the model discussed earlier. Individuals' payoffs are the same. The terrorist organization cares about efficacy, which is increasing in the quality of its operatives. Since each operative also imposes some costs, the organization might not be willing to take all comers.

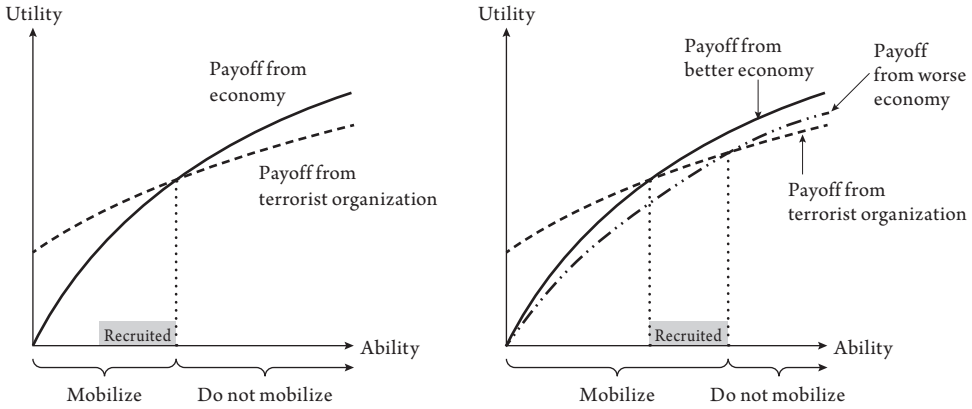

FIGURE 6.5. Terrorist mobilization decision.

The first cell of Figure 6.5 illustrates the equilibrium of this expanded model. Among those willing to mobilize, the organization accepts only individuals whose ability lies in the rectangle labeled "Recruited." These are the highest quality operatives available to the organization. So, in this model, terrorist operatives are not the lowest ability individuals. This agrees with the empirical finding that terrorist operatives are neither poor nor ill educated relative to the societies from which they are drawn.

In this model, the economy matters for terrorist mobilization and efficacy, even though terrorist operatives are not those with the least economic opportunity. A comparative static makes this clear. The second cell of Figure 6.5 shows what happens if the economy worsens (denoted by the dash-dot curve labeled "Payoff from worse economy"). The pool of potential recruits expands. As a result, the quality of terrorist operatives and the efficacy of the terrorist organization increase.

This theoretical argument is not sufficient to establish that the economic opportunity cost or screening mechanisms are at work. But it does offer a reinterpretation of the empirical evidence, showing that the data do not entail the conclusion that the motivation for mobilization must be unrelated to economic opportunity. That is, the data showing that terrorist operatives are neither poor nor ill educated is consistent both with a reasonable theoretical interpretation in which lack of economic opportunity does matter for mobilization and with a reasonable theoretical interpretation in which economic opportunity does not matter.

Unlike Krehbiel's reinterpretation that we discussed earlier, Bueno de Mesquita is not making an omitted-variables bias critique. Suppose we randomly assigned incomes in the population and then found that terrorist operatives were not poor relative to the population. This would still be a common implication of a model in which economic opportunity doesn't matter and of a model embodying the opportunity cost and screening mechanisms. After all, in the latter case, the terrorist organization would still select the highest quality operatives from among those who mobilized. The interpretive problem is not one of confounding.

This does not mean that different data could not be used to adjudicate between these two theoretical interpretations. To take the best case, suppose we had complete data on willingness to mobilize and ability. Bueno de Mesquita's argument implies that, among those willing to mobilize, recruitment should be positively related to ability. Krueger's argument would suggest no such relationship.

In practice, we are rarely in that kind of best case. To go further, we need to identify other empirical implications that differ between the two models, so that data might distinguish between them. In Chapter 8 we return to this and discuss empirical work that attempts to shed further light on this debate.

6.3 Elections and Natural Disasters

Students of voter behavior have long debated whether voters are up to the job given them in a democracy. The early literature on voter competence was concerned with whether voters were sufficiently informed to make good decisions (Campbell et al., 1960; Fair, 1978; Kinder and Sears, 1985; Popkin, 1991; Sniderman, Brody, and Tetlock, 1993; Lupia, 1994; Delli Carpini and Keeter, 1996). A more recent literature focuses on another aspect of voter competence—their rationality.

To assess whether voters behave rationally, that literature asks how election outcomes change in response to various shocks to voter welfare. One strand examines the electoral consequences of natural disasters and government responses to them (Abney and Hill, 1966; Achen and Bartels, 2004; Healy and Malhotra, 2010; Healy, Malhotra, and Mo, 2010; Bechtel and Hainmueller, 2011; Gasper and Reeves, 2011; Cole, Healy, and Werker, 2012; Huber, Hill, and Lenz, 2012; Chen, 2013). Another strand examines the electoral consequences of economic shocks (Ebeid and Rodden, 2006; Wolfers, 2002; Leigh, 2009; Kayser and Peress, 2012). Many such studies report that

incumbent electoral fortunes suffer following negative events outside the control of policymakers (Achen and Bartels, 2004; Wolfers, 2002; Leigh, 2009; Healy, Malhotra, and Mo, 2010). These results are typically interpreted as evidence of voter irrationality. The basic argument is that, if voters are rational, then they should not blame incumbents for events outside of the incumbents' control.

Ashworth, Bueno de Mesquita, and Friedenberg (2018) offer a reinterpretation, challenging the claim that such empirical findings imply that voters are not behaving rationally. They present a model in which incumbent electoral fortunes can suffer following a natural disaster or economic downturn, even though voters are rational and politicians exert no control over the occurrence of such events.

To see how the argument works, consider a model of the effect of shocks on electoral fortunes. Each politician has quality θ that can be high ($\overline{\theta}$) or low ($\underline{\theta}$). The incumbent is high quality with probability p_I and the challenger is high quality with probability $p_C < p_I$. The assumption that the incumbent is higher quality, in expectation, than the challenger is meant to represent things like learning on the job and the fact that the incumbent has already been positively selected by winning an earlier election. Rational voters prefer higher quality politicians because they are expected to provide better governance outcomes in the future.

Before the election, the voter observes two things—a random shock ω and a governance outcome. The shock is either small ($\underline{\omega}$) or large ($\overline{\omega}$), and its value is completely random, not influenced in any way by the incumbent.

The governance outcome g depends on both the incumbent's quality and the shock. Its values satisfy the following relations:

$$g(\overline{\theta}, \underline{\omega}) = g(\underline{\theta}, \underline{\omega}) > g(\overline{\theta}, \overline{\omega}) > g(\underline{\theta}, \overline{\omega}). \tag{6.1}$$

The equality says that the incumbent's quality is irrelevant for governance outcomes when the shock is small. The first inequality implies that a large value of the shock reduces the governance outcome, no matter the incumbent's quality. The second inequality says that, when the shock is large, high-quality incumbents produce better governance outcomes than do low-quality incumbents.

The shock represents a natural disaster, such as a hurricane or the emergence of a novel virus. A hurricane is an act of God, not something the incumbent can influence. This is represented by the assumption that the value of the shock is independent of the incumbent's quality. Moreover, hurricanes

are destructive, so damage (similar to governance outcomes) is avoided when storms do not occur. But when storms do land, the damage depends on infrastructure investment, emergency preparedness and response, and so on. The incumbent's quality is meant to represent the capacity and willingness to engage in such preparations.

The inequalities in Condition 6.1 have implications for voter learning. The size of the shock affects how much information the voter has about the incumbent. The voter perfectly learns the incumbent's quality if there is a large shock, but the voter learns nothing if there is a small shock.

This kind of assumption about informativeness is plausibly representational. Suppose better incumbents do a good job preparing for storms than do worse incumbents. If the weather is calm, the voter does not see the difference in preparedness. If a hurricane comes, the voter does see the difference in preparedness, because good preparation leads to only minor damage. Of course, the assumption made here is too strong. It is surely not true that voters learn nothing about the incumbent in the absence of a disaster and learn everything about the incumbent in the presence of a disaster. Ashworth, Bueno de Mesquita, and Friedenberg show, in their paper, that all that is needed is that the voter learns more when there is a disaster than when there is not. We make the stronger assumption here just to ease exposition.

Because the voter wants to elect whichever candidate is most likely to be high quality, the first step in solving the model is to determine what the voter believes about the incumbent, based on what the voter observes. If there is a small shock, the voter learns nothing and continues to assess probability p_I that the incumbent is high quality. We record this in the top row of Table 6.1 on the left. If there is a large shock, the voter observes either $g(\overline{\theta}, \overline{\omega})$ or $g(\underline{\theta}, \overline{\omega})$, depending on the true quality of the incumbent. The better outcome reveals that the incumbent is high quality, while the worse outcome reveals that the incumbent is low quality. We record this in the bottom row of Table 6.1 on the left.

Next, we determine how the voter votes. Because $p_I > p_C$, when there is a small shock and the voter learns nothing new, she continues to believe the incumbent is more likely to be high quality than is the challenger. As a result, the voter reelects the incumbent whenever there is a small shock. If there is a large shock, the voter learns the incumbent's quality from her observation of damage and preparedness. If that quality turns out indeed to be high, the incumbent wins reelection. But if that quality turns out to be low, the incumbent loses. We record these electoral outcomes in Table 6.1, right side.

TABLE 6.1. The Voter's Posterior Probability Assessment That the Incumbent Is High Quality and the Incumbent's Electoral Fortunes, as a Function of the Incumbent's Actual Quality and the Size of the Shock.

(a) Posterior probability			(b) Electoral fortunes		
	High quality	Low quality		High quality	Low quality
Small shock	p_I	p_I	Small shock	Reelected	Reelected
Large shock	1	0	Large shock	Reelected	Defeated

Because there is a positive probability that the incumbent is low quality, there is a positive probability that the incumbent is defeated following a large shock. But the incumbent is reelected for sure if there is a small shock. Hence, by providing the voter with new information, the large shock reduces the ex ante probability the incumbent is reelected, even though the shock itself is not the incumbent's fault.[2]

This result is quite stark and, as stated, implausible as a representation of the world. It is clearly not the case that incumbents win reelection for certain when there is no disaster. But, as we've already mentioned, the strong informational assumption that gives rise to this stark result on reelection rates is not necessary for the key implication that the incumbent's probability of reelection is higher when there is no disaster. For instance, Ashworth, Bueno de Mesquita, and Friedenberg show that in nearby models in which governance outcomes depend on other, unobservable factors in addition to natural disasters or incumbent quality, it is still true that natural disasters affect electoral fortunes, but it is no longer true that incumbents win for certain absent a disaster.

In this model, there is a negative correlation between the exogenous shock and incumbent electoral fortunes. To relate this to the empirical literature, consider a large population of locations, each of which is represented by the model just sketched. Some of these locations have a hurricane and some do not. Since hurricanes happen by chance, the distribution of high- and low-quality incumbents is the same in each subpopulation. The model's implication is that incumbents win at a higher rate in locations that do not have hurricanes than in locations that do. Hence, the difference in electoral fortunes is a common implication of this rational-voter model and the irrational-voter

2. If the voter started with the belief that the incumbent was less likely to be high quality than the challenger, then the effect of the shock would be reversed.

story offered in the prior literature.[3] As such, this reinterpretation does not require an interlocutor to conclude that voters are rational; it merely shows that the evidence on offer does not entail the conclusion that voters are irrational.

6.4 Gleanings

We've just looked at three debates in three different literatures. We deliberately chose examples in which the associated theoretical models were relatively accessible and simple to present. But a growing body of theoretical research has taken up the challenge of reinterpretation on topics ranging from media bias (Wolton, 2019) to electoral violence (Harish and Little, 2017) to lobbying (Schnakenberg, 2017).

None of the papers we discussed is anything like the last word on these substantive topics. It is always a good bet that scholars will come up with new evidence and new reinterpretations. That doesn't make it a mistake to offer an interpretation or think about the implications of a particular mechanism. Doing so, even knowing the risk of a future reinterpretation, is an important step in the accumulation of knowledge.

The particular works we discussed highlight two general points that are worth expanding on.

First, as we noted in our discussion of party effects in Congress, reinterpretation sometimes takes the form of an omitted-variables bias critique that speaks to commensurability. Krehbiel, for instance, uses theory to show that earlier empirical work failed to account for ideological sorting when comparing Republicans and Democrats. The objection would evaporate if we could randomly assign members to parties. If we still saw systematic differences in behavior, they could not be due to ideological sorting. So a difference in behavior between Democrats and Republicans would not be a common

3. It is worth noting that this reinterpretation was built on the assumption that natural disasters, while random, have consequences that interact with policy. While that is true for many of the shocks used in the empirical literature, it is not the case in two prominent papers, one using shark attacks and the other college football losses (Achen and Bartels, 2004; Healy, Malhotra, and Mo, 2010). These findings, then, would constitute the most compelling evidence for the voter irrationality mechanism, as they are immune from the sort of reinterpretation just offered. However, careful empirical scrutiny calls into question the empirical findings in both papers (Fowler and Montagnes, 2015; Fowler and Hall, 2018), as we will discuss in Chapter 7.

implication of the party pressure mechanism and the ideological sorting mechanism under random reassignment.

But not all reinterpretations are of this form. Indeed, neither of the other two reinterpretations discussed in this chapter are critiques about omitted variables. We have already discussed this in the context of terrorist recruitment. It is even clearer in the case of disasters and elections. There, the appeal of the initial empirical papers was precisely that disasters are as-if randomly assigned. The reinterpretation shows that the meaning of a correlation between as-if random disasters and incumbent reelection is not as obvious as was initially thought.

Second, you don't necessarily need to invent a brand new mechanism to reinterpret. Reinterpretation often involves making new use of mechanisms that are already well understood. In the case of the literature on party effects in Congress, Krehbiel invokes the canonical model of ideological voting. His contribution is not creating this model, it is seeing its implications for the interpretation of a set of empirical results. Ashworth, Bueno de Mesquita, and Friedenberg's Bayesian learning model similarly builds on canonical models of voter learning due to Achen (1992) and Bartels (1993). The screening model of terrorist organizations is not standard in that literature. But it too invokes standard models of previously understood mechanisms—in this case, a model of positive selection—applying it in a new context.

Each of these illustrates a point that harkens back to our discussion of well-understood mechanisms in Chapter 4. Reinterpreting is about pointing out that models embodying different plausible mechanisms have common implications. You are more likely to spot a reinterpretation the more mechanisms you understand. This is not to say it is enough to simply pick any model from the theorist's toolkit that generates a common implication. Your new mechanisms must have a plausible claim of similarity to the target. Evaluating similarity is in large part a matter of substantive knowledge of the phenomenon in question. Hence, being good at reinterpreting depends both on being familiar with how to model many mechanisms and knowing enough substance to judge their applicability in particular settings. Finally, those pieces of background knowledge must be accompanied by the habit of always asking whether empirical findings can be understood through alternative, substantively plausible mechanisms other than those already on offer in a literature.